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by

Chanont Banternghansa, Athiphat Muthitacharoen, Archawa Paweenawat and Krislert Samphantharak

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Tax Incentives to Appear Small: Evidence from Thai Firms and Corporate Groups*

Chanont Banternghansa† Athiphat Muthitacharoen‡ Archawa Paweenawat§

Krislert Samphantharak¶

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Abstract

This paper studies the effects of SME tax incentives on firm behaviors. We use firm-level panel data of all registered firms in Thailand to analyze the effects of a large reduction in corporate income tax rates for SMEs in 2011. First, we find that firms responded strongly to the SME tax incentive as indicated by a sharp bunching of firms just below the threshold after the incentive was introduced. The responses were concentrated among firms with positive EBIT, implying a financial motive for firms to remain small. Second, the bunching was prominent for stand-alone firms, where we observe slower revenue growth for those below the threshold. Third, we do not observe bunching for corporate-group firms, but we find evidence of tax-motivated profit shifting among them instead, especially among firms in small groups with weak corporate governance. Our analysis suggests that transfer pricing was likely a primary channel. Finally, despite the unintended consequences, we find that the incentive significantly raised the probability of firm's survival and encouraged new firm registration, as the policy intended.

Keywords: bunching, tax incentives, business group, corporate tax, size-dependent policy, tax evasion

JEL Classifications: F23, H25, H26, K34, M42

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†Analysis Group: c.banternghansa@gmail.com

‡Faculty of Economics, Chulalongkorn University (Thailand): athiphat@gmail.com

§Puey Ungphakorn Institute for Economic Research (PIER), Bank of Thailand: ArchawaP@bot.or.th

¶University of California San Diego and PIER: krislert@ucsd.edu

1 Introduction

Many countries have supported small and medium enterprises (SMEs) using size-based tax incentives (Bergner, Bräutigam, Evers, and Spengel, 2017). While these SME tax incentives are often used to mitigate tax costs and financial constraints and to promote startups and employment, they incur tax expenditure, create tax compliance problem, and potentially generate disincentives for firms to invest and grow. Firms with different ownership structures (e.g., stand-alone firms and those in corporate groups) may also respond differently to such size-based tax treatments as their abilities to benefit from the policy are not the same. Despite the widespread adoption of SME tax incentives, there is little micro-level evidence on their causal impacts on firm behaviors.

This paper investigates how firms respond to a size-dependent tax incentive and how the responses may differ by ownership structures. We take advantage of a significant reduction of corporate income tax rates for SMEs in Thailand in 2011. Our identification strategy is based on a bunching framework (Kleven and Waseem, 2013) and a difference-in-difference approach before and after the introduction of the tax incentive. By granting a low-tax scheme for firms with total revenue not over 30 million baht, the government has introduced an important tax notch in the corporate income tax system.[1]

We use administrative data containing annual financial statements from 2004 to 2017 for the universe of firms in Thailand. In addition, we use corporate shareholding data to identify corporate groups, defined as a collection of firms that belong to the same owners. Our newly constructed dataset therefore allows us to study the effects of the SME tax incentive at both firm and corporate group levels.

We document four sets of empirical findings. First, we find that firms respond strongly to the SME tax incentive as indicated by sharp bunching just below the threshold that did not exist before the introduction of the tax incentive. We find that the bunching response is mainly driven by firms with positive EBIT and there is no bunching among firms with negative and zero EBIT where there is no incentive to bunch. The bunching response is also higher in the retail sector where there is relatively larger opportunity to manipulate or misreport revenue.

Second, our difference-in-difference analyses illustrate that the tax incentive produces negative effects on revenue growth for stand-alone firms. We find that following the 2011

[1] The exchange rate for Thai baht ranged between 26.8 and 41.6 baht per US dollar during 2004–2017. The average rate was 34.3.

introduction, the revenue growth for stand-alone firms below the SME threshold declines more significantly than those above the threshold.

Third, we find evidence of tax-motivated profit shifting among corporate-group firms. Our difference-in-difference analyses for corporate-group firms with both SMEs and non-SMEs show that, following 2011, the profit growth of SMEs increases significantly relative to that of non-SMEs. We also find that transfer pricing is likely a primary channel for the profit shifting among corporate groups. Such profit shifting is concentrated among firms in small corporate groups that do not have any firms listed in the stock exchange (hence likely to have weaker corporate governance). This finding underlines the importance of tax avoidance through domestic profit shifting and the role of corporate governance, which are crucial for countries that allow preferential tax schemes for certain types of firms. These issues have so far received little attention in the literature compared to cross-border transfer pricing.

Finally, while our findings indicate that there is a strategic response to the tax notch, the tax incentive seems to help relax financial constraints faced by SMEs and yield intended positive impacts. Specifically, we find that the incentive significantly enhanced the survival probability of existing firms. We also document some positive effects on the formalization of new firms, with higher growth rates of firm registration among sectors that likely benefit more from the tax-reduction scheme.

This paper is directly related to the growing pool of studies that have documented how firms respond to thresholds or notches created by tax system. Examples include bunching to avoid complying to the value added tax regulations (Harju, Matikka, and Rauhanen, 2019; Liu, Lockwood, Almunia, and Tam, 2019), and to stay below the enforcement radar (Almunia and Lopez-Rodriguez, 2018). We contribute to this literature in two important ways. First, we emphasize the heterogeneity in response to size-dependent regulation with respect to corporate structure (stand-alone vs. corporate group). Second, we provide credible estimates of how the SME tax incentive causes stand-alone firms to limit their growth and encourages tax-motivated profit shifting among corporate-group firms.

The remainder of this paper is organized as follows. Section 2 discusses related existing studies and the contribution of this study to the literature. Section 3 provides institutional background and describes the data used in the analysis of this study. Sections 4–7 discuss the empirical strategy and results for each analysis: (1) bunching, (2) revenue growth of stand-alone firms, (3) profit shifting of firms in corporate groups, and (4) survival and formation of firms.

Finally, Section 8 concludes the study and discusses policy implications.

2 Related Literature

This study is related to several strands of literature. First, our paper joins a growing number of studies that investigate how eligibility thresholds for certain taxes or tax enforcements affect firm evasion and avoidance responses. Recent studies have documented bunching by small firms at the revenue-based registration thresholds for value added tax (Harju, Matikka, and Rauhanen, 2019; Liu, Lockwood, Almunia, and Tam, 2019). Such firm bunching response is not limited to VAT thresholds. Bachas and Soto (2018) shows that Costa Rican firms bunch with respect to notches created by the corporate income tax system. Almunia and Lopez-Rodriguez (2018) also finds that large Spanish firms bunch below a revenue-based threshold to avoid entering the stricter tax enforcement regime. We contribute to this literature by illustrating the heterogeneity of the response to the size-dependent tax regime with respect to corporate structure.

Another strand of related literature consists of studies on how regulations that depend on firm size affect growth and productivity. For example, Garicano, Lelarge, and Van Reenen (2016) shows that France's size-dependent labor regulations have important effects on the productivity distribution. Benedek, Budina, Deb, Gracia, Saksonovs, and Shabunina (2017) finds that size-related tax incentives that do not specifically target R&D investment have a significant negative relationship with firm productivity and growth. In this paper, we extend this literature by providing an estimate of the impact of the tax incentive for SMEs on firm growth in a developing country.

Finally, our findings are consistent with the insight from studies that investigate the base erosion and profit-shifting issues for developing countries. Examples include Fuest, Hebous, and Riedel (2011), Crivelli, de Mooij, and Keen (2016), Johannesen, Tørsløv, and Wier (2020), and Muthitacharoen and Samphantharak (2020). We contribute to this literature by illustrating that the tax avoidance through profit shifting issue is not limited to multinational firms but is also likely to be important for domestic corporate groups.

3 Background and Data

All Thai firms are subject to corporate income taxation which accounts for approximately 26% of the country's total tax revenue in 2020. Generally, they are subject to a flat tax rate on their net profit. Since 2011, the government has introduced a preferential tax scheme for firms with total revenue not over 30 million baht[2] and registered capital not over five million baht. The preferential scheme includes relatively low corporate income tax rates based on net profit. Its objectives were to ease SMEs' financial constraints and to promote their competitiveness. This has created an important tax notch in the corporate income tax system. Table 1 shows Thailand's corporate income tax rates for general firms and SMEs.

Table 1: Corporate income tax rates: General and SMEs (%)

Year	SMEs					General
	Size of Net Profits					
	0–150k	150k–300k	300k–1M	1M–3M	>3M	
2004	30	30	30	30	30	30
2005	30	30	30	30	30	30
2006	30	30	30	30	30	30
2007	30	30	30	30	30	30
2008	0	15	15	25	30	30
2009	30	30	30	30	30	30
2010	30	30	30	30	30	30
2011	0	15	15	25	30	30
2012	0	15	15	23	23	23
2013	0	0	15	20	20	20
2014	0	0	15	20	20	20
2015	0	0	10	10	10	20
2016	0	0	10	10	10	20
2017	0	0	15	15	20	20
2018	0	0	15	15	20	20

Note: In 2008, firms are considered SMEs and qualify for the reduced tax rates if having registered capital not over 5 million baht. Since 2011, SMEs must have registered capital not over 5 million baht and have revenue not over 30 million baht.

This study analyzes a panel of firms in Thailand from 2004 to 2017 using two datasets. The first dataset is from the Corporate Profiles and Financial Statements (CPFS) database of

[2]The scheme also requires that a firm must never have annual revenue above 30 million baht in any year from 2011 to the respective year.

the Department of Business Development (DBD) at the Ministry of Commerce. All registered firms in Thailand are legally required to submit their annual financial statements to the DBD. The CPFS database includes the value of firms' assets, liabilities, revenues, and expenses. The database also provides firms' information such as registration year, registration type, status, and primary industry.[3]

The second dataset, also from the DBD, is a snapshot of corporate ownership as of 2017. This dataset contains information of each owner including name, type, nationality, and the number of shares owned.[4] In this paper, we focus on corporate shareholding and construct ownership networks based on the information from this dataset.

Specifically, as in Apaitan, Banternghansa, Paweenawat, and Samphantharak (2020), if firm A owns shares of stock in firm B, we consider that firms A and B belong to the same ownership network or the same corporate group. Around 5.4 percent of registered firms in Thailand belong to corporate groups. While more than half of the corporate groups consist of only two firms, there are some exceptionally large corporate groups. For example, the biggest corporate group in Thailand consists of almost 400 firms. In terms of industry diversification, corporate groups are rather concentrated. More than half of corporate groups consists of only firms in the same industry.[5]

Table 2 shows summary statistics for each analysis performed later in this study: the main bunching analysis in Section 4 (Panel A), the difference-in-difference analyses of stand-alone firms in Section 5 (Panel B), the difference-in-difference analysis of corporate-group firms in Section 6 (Panel C), as well as the analysis of firm's survival and formation in Section 7 (Panels D and E).

4 Bunching Analysis

4.1 Measuring Bunching

Following Kleven and Waseem (2013), we measure bunching at the SME tax notch by comparing the actual distribution to the counterfactual distribution in the absence of the tax notch. This

[3]We exclude holding companies from our analysis as they do not directly engage in production activities. For more details, see Banternghansa, Paweenawat, and Samphantharak (2019).

[4]Owners could be ordinary persons or juristic persons, including other domestic firms, foreign firms, investment funds, or public or state agencies.

[5]For more information about Thai corporate groups, see Apaitan, Banternghansa, Paweenawat, and Samphantharak (2020) and Banternghansa and Samphantharak (2021).

Table 2: Descriptive statistics

	N	Mean	Median	S.D.
Panel A: Main bunching analysis				
Revenue	599,627	26,160,208	24,442,722	8,322,355
Panel B: Stand-alone firms analysis (Difference-in-Difference analysis)				
Revenue growth	614,466	0.043	0.000	0.553
Treat	614,466	0.723	1.000	0.447
Post	614,466	0.462	0.000	0.499
Lagged total assets (log)	614,354	16.287	16.262	1.043
Panel C: Corporate-group firms analysis (Difference-in-Difference analysis)				
Profit growth	44,222	-0.323	-0.023	1.266
Treat	44,222	0.318	0.000	0.466
Post	44,222	0.377	0.000	0.485
Lagged total assets (log)	44,222	18.629	18.673	2.182
Revenue growth	97,597	0.163	0.011	0.575
Input costs growth	38,915	0.090	0.016	0.589
Debt growth	29,514	0.102	0.000	0.862
Panel D: Survival analysis (Difference-in-Difference analysis)				
Survival	469,652	0.977	1.000	0.151
Treat	469,652	0.731	1.000	0.443
Post	469,652	0.507	1.000	0.500
Lagged total assets (log)	469,649	16.236	16.171	1.092
Lagged current assets (log)	469,596	15.551	15.654	1.244
Firm age	469,652	12.036	10.000	9.069
Panel E: New firm formation (Difference-in-Difference analysis)				
Registration growth	2,962	0.203	0.000	0.785
Treat	2,962	0.887	1.000	0.317
Post	2,962	0.548	1.000	0.498

Note: In 2008, firms are considered SMEs and qualify for the reduced tax rates if having registered capital not over 5 million baht. Since 2011, SMEs must have registered capital not over 5 million baht and have revenue not over 30 million baht.

can be written as

$$b = \frac{\sum_{j=y_L}^{y^*}(c_j - \hat{c}_j)}{(\sum_{j=y_L}^{y^*}\hat{c}_j)/N}, \tag{1}$$

where y^* is the SME threshold, y_L is the lower limit of the excluded region, c_j is actual number of firms in each revenue bin (width of 100,000 baht) and $\hat{c}_j$ is the counterfactual number in each revenue bin in the absence of the tax notch, and N is the number of bins within the interval $[y_L, y^*]$. The bunching parameter b reflects the size of excess bunching relative to the average height of the counterfactual distribution to the left of the notch.

The counterfactual distribution of reported revenue is estimated by fitting a fifth-order polynomial equation and excluding the areas around the notch where bunching occurs.[6] The equation can be written as

$$c_j = \sum_{i=0}^{p} \beta_i (z_j)^i + \sum_{i=z_L}^{z_U} \gamma_i \mathbf{1}\left[z_j = i\right] + \nu_j, \tag{2}$$

where c_j denotes number of firms in bin j, z_j denotes revenue level of bin j, p denotes polynomial order, $[z_L, z_U]$ denotes the excluded region, and ν_j denotes the error term. The predicted value ($\hat{c}_j$) from this equation represents the counterfactual distribution that is used in equation (1). Note that the lower limit of the excluded region (y_L) is set where the bunching begins, and the upper limit is estimated in an iterative procedure to ensure that the excess mass below the notch equals the missing mass above.

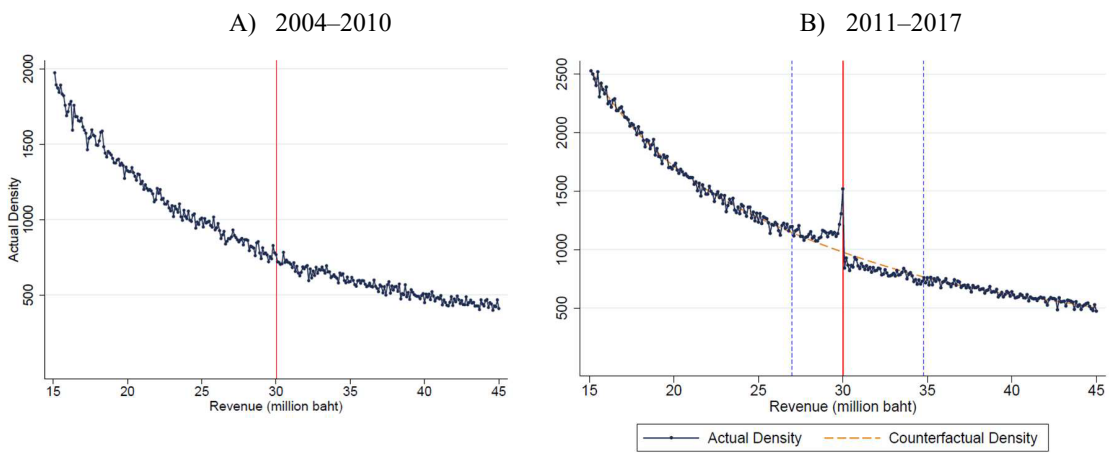
4.2 Bunching at the Tax Notch

To determine whether firms respond to the SME tax notch, Figure 1 shows histograms of revenue around the SME threshold with the bin width of 100,000 baht. There is a sharp bunching just below the 30-million-baht threshold during 2011–2017 and that bunching mass did not exist before the introduction of the tax incentive in 2011. These findings suggest that firms respond strongly to the SME tax incentive.

Table 3 reports the bunching estimates. The overall bunching estimate is 2.62, i.e., the total excess bunching mass is approximately 2.6 times the average height of the counterfactual over the excluded range. Across sectors, the bunching is largest in the retail trade sector, as shown in Table 3. One explanation is that retail businesses might find it relatively easy to

[6] We also estimate the polynomial equations of the 4th, 6th, and 7th order. The results are consistent and are available upon request.

Figure 1: Histograms of revenue around the SME threshold



Note: This figure shows the histograms of firms' revenue by pooling data of all firms from 2004 to 2010 (panel A) and from 2011 to 2017 (panel B). The bin width is 100,000 baht. The red vertical line denotes the SME threshold of 30 million baht. The blue vertical dashed lines denote lower bound and upper bound of the excluded region (27–34.8 million baht). The orange dashed line is counterfactual density fitted by excluding bins around the SME notch.
Source: Authors' estimate.

misreport sales because the VAT self-enforcement mechanism breaks down at the retail stage.[7]

Table 3: Bunching estimates by major sectors

	Bunching Estimate	Standard Error
All sectors	2.622	0.015
Manufacturing	2.747	0.032
Services	2.102	0.021
Wholesale	2.363	0.029
Retail	3.669	0.076

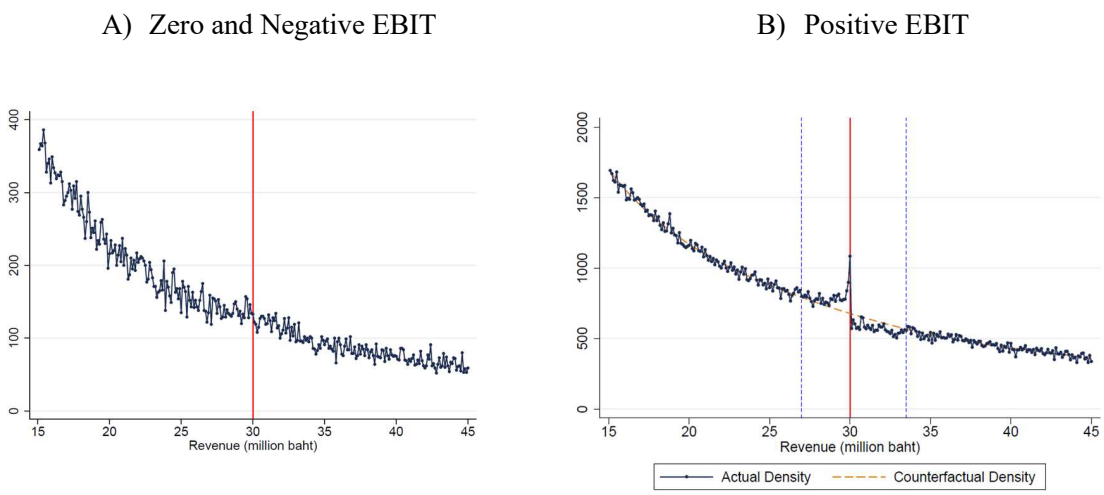
Note: This table shows the bunching estimates around the SME threshold for 2011–2017. The estimation methodology is based on Kleven and Waseem (2013).

We also find that the bunching response is mainly driven by firms with positive EBIT as shown in Figure 2. There is no bunching among firms with negative and zero EBIT where there is no incentive to bunch. This result suggests that the response is consistent with the monetary incentive.

What we observe is likely to be a mix of the real operation response and the under-reporting response. We are not able to precisely decompose the contribution of each mechanism.

[7]See, for example, Pomeranz (2015) and Naritomi (2019).

Figure 2: Histogram of revenue around the SME threshold by profitability



Note: This figure shows the histogram of firms' revenue by pooling data of all firms from 2011 to 2017 by profitability. The bin width is 100,000 baht. The red vertical line denotes the SME threshold of 30 million baht. The blue vertical dashed lines denote lower bound and upper bound of the excluded region. The orange dashed line is counterfactual density fitted by excluding bins around the SME notch.
Source: Authors' estimate.

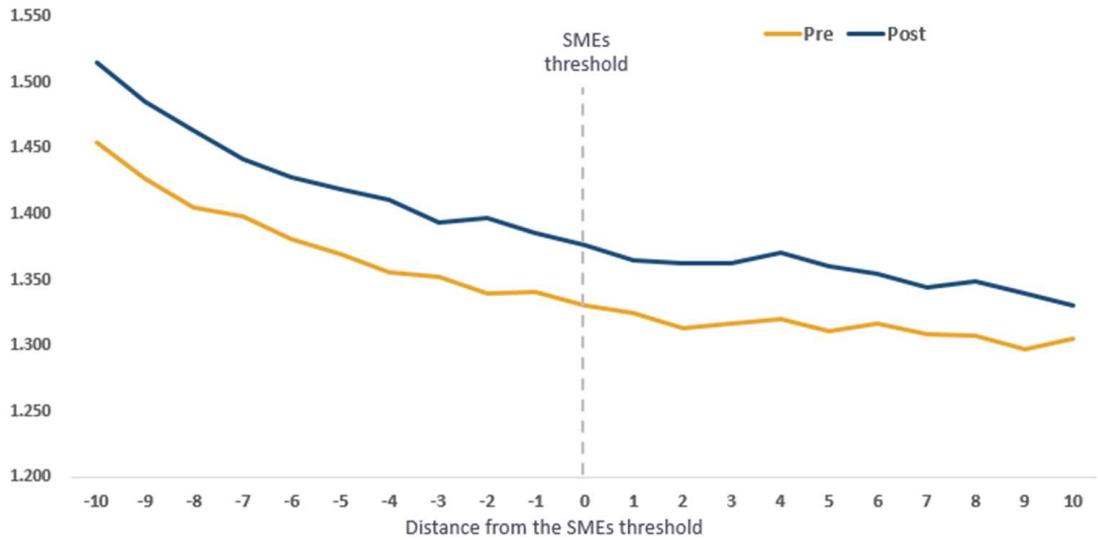
However, using the information on firms' input costs, we can provide some suggestive evidence that a significant part of the observed bunching could be attributed to the real response.

In case where the revenue bunching is due to the real operation response, we expect firms to lower their variable inputs in proportion. Consequently, the distribution of the ratio between revenue and variable costs is likely to be smooth around the SME notch. On the other hand, if the bunching is not from the changes in real operation, the ratio may not be smooth because input costs are generally deductible for corporate income tax, making firms reluctant to reduce their reported costs. If most firms under-reported revenue but did not adjust their reported costs accordingly, we would expect to see a drop in the average revenue-cost ratio just below the tax notch relative to that above the notch.

Figure 3 illustrates the average revenue-cost ratio of SMEs in bins of 2 million baht on both sides of the threshold. The revenue-cost ratio is defined as total revenue divided by variable costs of goods and services sold (COGS). Such costs include salary expense which is relatively difficult to misreport due to withholding taxes. The evidence of smooth distribution of the revenue-cost ratio around the notch suggests that a significant part of the effect could be attributed to the real operation response.

It is important to note that this finding provides only a suggestive evidence since some

Figure 3: Revenue-cost ratio around the SMEs threshold



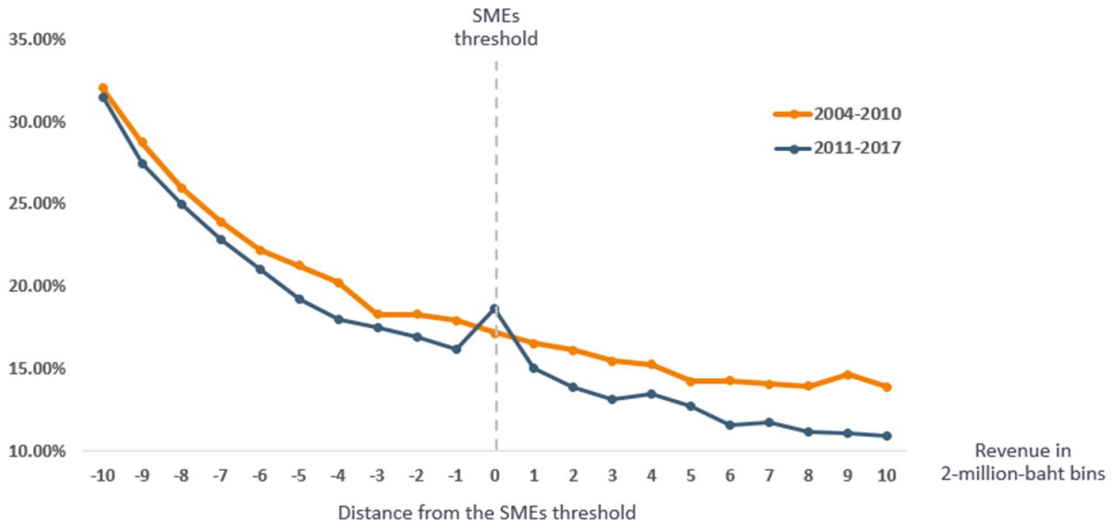
Note: This figure shows the average revenue-input ratio of SMEs in bins of 2 million baht on both sides of the threshold. The revenue-input ratio is defined as total revenue divided by costs of all goods and services.
Source: Authors' estimate.

firms may choose to reduce their reported cost in proportion to their under-reported revenue. Although the cost is tax-deductible, lowering cost in proportion to sales may help avoid potential scrutiny from the tax authority.

4.3 Persistence of Bunching

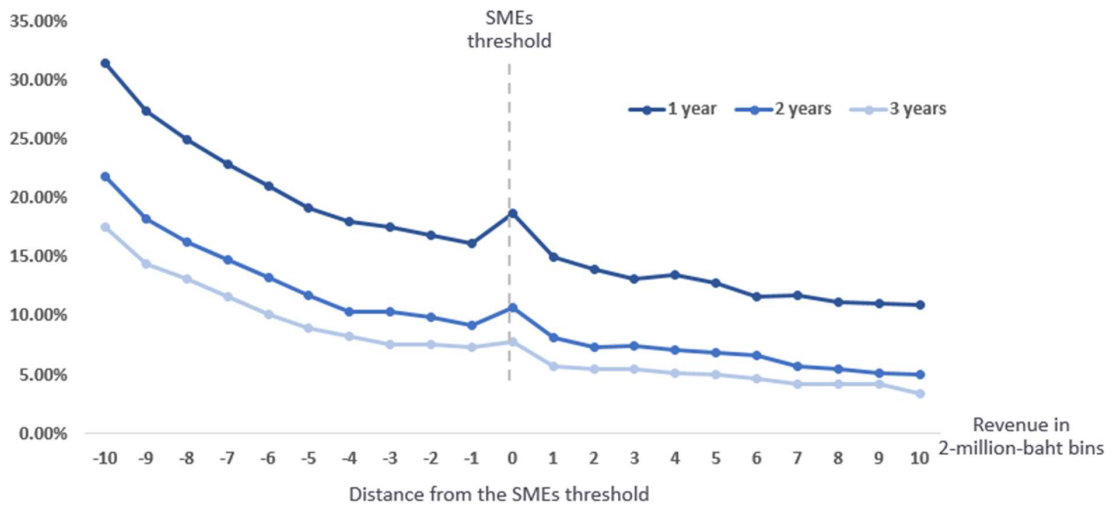
We also explore how the SME threshold may affect firm behaviors over time. Figure 4 shows the one-year persistence rates of SMEs in 2-million-baht bins on both sides of the threshold before and after the introduction of the SME tax incentive. The persistence rate is defined as the proportion of firms that remained in the same revenue bin from one year to the next. As illustrated in the figure, during 2011–2017, the persistence rate in the bin just below the threshold (2.8–3 million baht) is noticeably larger than that in other bins around the threshold. This pattern is not observed before the SME tax incentive introduction. We also find that this pattern remains after several years, as shown in Figure 5. This finding implies that firms attempted to stay below the tax notch for multiple years.

Figure 4: Persistence rate at the SME threshold: 1 year (2004–2010 vs. 2011–2017)



Note: This figure shows the persistence rate, defined as the probability that a firm remains in the same 2-million-baht revenue bin from one year to another.
Source: Authors' estimate.

Figure 5: Persistence rate at the SME threshold: 1, 2, and 3 years (2011–2017)



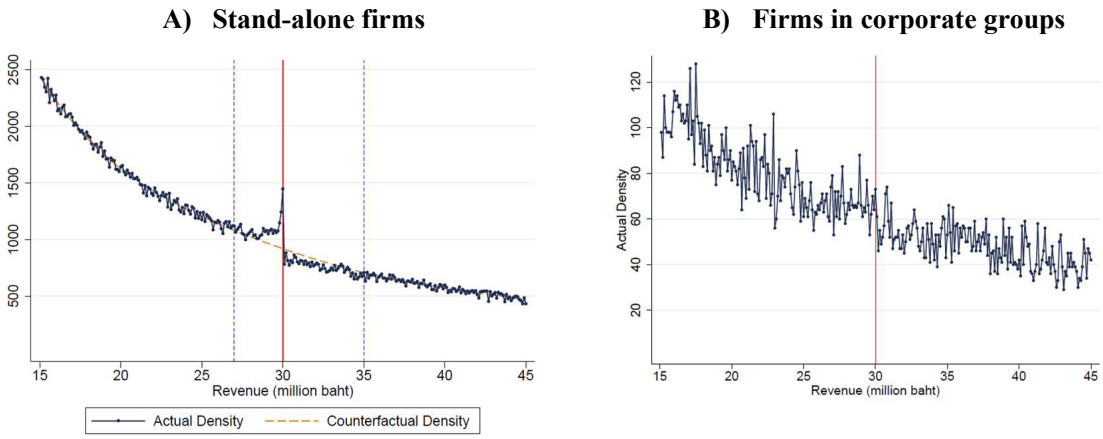
Note: This figure shows the persistence rate, defined as the probability that a firm remains in the same 2-million-baht revenue bin from one year to another.
Source: Authors' estimate.

4.4 Heterogeneity Analysis: Stand-Alone vs Corporate-Group Firms

Finally, we perform heterogeneity analyses with respect to corporate ownership structure. We divide firms into two groups: stand-alone firms and corporate-group firms. Figure 6 shows that

the bunching responses are primarily driven by stand-alone firms. For firms in corporate groups, we do not find bunching evidence. This finding suggests that these two types of firms are likely to respond differently to the SME tax incentive. One possible explanation is that stand-alone firms have relatively limited opportunity to collusively shift revenue or profits to other firms. Some stand-alone firms thus rely on constraining (reported) revenue in order to benefit from the SME tax scheme. We further investigate the differences in responses between these two types of firms in the next two sections.

Figure 6: Histogram of revenue around the SME threshold by ownership structure



Note: This figure shows the histogram of firms' revenue by pooling data of all firms from 2011 to 2017 by ownership structure. The bin width is 100,000 baht. The red line denotes the threshold of 30 million baht. The blue vertical dashed lines denote lower bound and upper bound of the excluded region. The orange dashed line is counterfactual density fitted by excluding bins around the SME notch.
Source: Authors' estimate.

5 Growth of Stand-Alone Firms

Having established that the SME tax incentive seems to affect reported revenue of stand-alone firms, we further analyze how the size-dependent regulation has affected the growth of stand-alone entrepreneurs over time. In this section, we limit the sample to stand-alone firms and perform the difference-in-difference analyses.

We define our treatment group to be firms that satisfied the following conditions: (1) have lagged revenue in the range of 10–30 million baht and (2) have never had revenue above 30 million baht in the period from 2011 to the respective year.[8] Control group includes firms

[8]Note that we do not impose a requirement on registered capital since the information is only available for 2017. However, over 85% of all firms in that year have registered capital not exceeding 5 million baht.

with lagged revenue in the range of 30–50 million baht. Post includes years from 2011 onwards. The equation can be written as:

$$y_{it} = \alpha_1 Treat_{it} \cdot Post_t + \alpha_2 Treat_{it} + \alpha_3 Post_t + \delta_i + \epsilon_{it} \tag{3}$$

where y_{it} denotes the revenue growth of firm i in year t, $Treat_{it}$ denotes a dummy variable that equals one for treated firms and zero for controlled firms, $Post_t$ denotes a dummy variable that equals one for the years including and after 2011 and equals zero otherwise, and δ_i denotes a dummy variable for firm i. Under the identification assumption that unobserved determinants of revenue growth (ϵ_{it}) do not change differentially on average across the treatment and control groups around the reform, the coefficient α_1 represents the causal effect of the SME tax policy on the revenue growth. The estimation period ranges from 2004 to 2017.

Table 4: Effects of the SME tax incentive on revenue growth of stand-alone firms

$RevGrowth_{it}$	(1) All	(2) All	(3) Positive EBIT	(4) Negative EBIT
$Treat_{it} \cdot Post_t$	-0.097^{***}	-0.098^{***}	-0.117^{***}	-0.042^{***}
	(0.003)	(0.003)	(0.004)	(0.010)
$Treat_{it}$	0.274^{***}	0.269^{***}	0.293^{***}	0.258^{***}
	(0.003)	(0.003)	(0.004)	(0.008)
$Post_t$	-0.012^{***}	0.381	0.289	-0.291^{**}
	(0.003)	(0.452)	(0.437)	(0.128)
Control	No	Yes	Yes	Yes
Sector-Year FE	No	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
R-squared	0.040	0.062	0.065	0.150
Observations	614,466	614,354	498,797	115,557
Number of firms	114,721	114,711	105,471	59,543

Note: This table presents the estimated impacts of the SME tax incentive on revenue growth of stand-alone firms. $Treat_{it}$ is a dummy variable that equals one for firms that (1) have lagged revenue in the range of 10–30 million baht and (2) have never had revenue above 30 million baht in the period from 2011 to the respective year, and equals zero for firms with lagged revenue above 30 million baht but not over 50 million baht. $Post_t$ is a dummy variable that equals one for the years including and after 2011. The control variable is lagged value of total assets in log form. Standard errors are heteroscedasticity-robust and clustered at firm level. Numbers in parentheses indicate standard error. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.
Source: Authors' estimate.

The results from our difference-in-difference analyses are reported in Table 4. Column (1) reports the result from baseline specification. For the specification in Column (2), we include

lagged value of total assets in log form, as a control variable, and industry-year fixed effects to capture sector-specific macroeconomic conditions. The results illustrate that the tax incentive produced adverse effects on the growth of stand-alone firms. As shown in Columns (1) and (2), after the introduction of the tax incentive in 2011, the revenue growth of stand-alone firms below the SME threshold declined by approximately 10 percentage points relative to the revenue growth of those above the threshold.

We also find that the negative impact on revenue growth is much higher among firms with positive EBIT. Among firms with positive EBIT, the revenue growth of those below the SME threshold declined by 11.7 percentage points relative to the revenue growth of those above the threshold (Column 3). For firms with negative EBIT, and hence no immediate tax benefit from staying below the threshold, the size of the interaction coefficient is much smaller in absolute term (Column 4).

6 Profit Shifting of Firms in Corporate Groups

Given that we do not find evidence of bunching among firms in corporate groups, we analyse further whether corporate-group firms take advantage of the SME tax incentives through other mechanisms by analysing the potential shifting of profits from non-SME firms to SME firms in the same corporate group. In this section, we therefore restrict the sample to corporate groups with SME firms.

We define our treatment group to be firms that satisfy the following conditions: (1) have lagged revenue not over 30 million baht and (2) have never had revenue above 30 million baht in the period from 2011 to the respective year. Control group includes firms with lagged revenue above 30 million baht. The estimating equation can be written as

$$y_{it} = \alpha_1 Treat_{it} \cdot Post_t + \alpha_2 Treat_{it} + \alpha_3 Post_t + \delta_i + \epsilon_{it} \tag{4}$$

where y_{it} denotes profit growth, revenue growth, debt growth, or input cost growth, $Treat_{it}$ denotes a dummy variable that equals one for treated firms and equals zero for controlled firms, $Post_t$ denotes a dummy variable that equals one for the years including and after 2011 and equals zero otherwise, and δ_i denotes a dummy variable for firm i. The estimation period ranges from 2004 to 2017.

We initially investigate the revenue growth before and after the introduction of the tax

incentive. Columns 1 and 2 of Table 5 show our estimation of Equation (4) with revenue growth as a dependent variable. We do not observe any significant effect on revenue growth, confirming what we find in the bunching analysis in Section 5.

However, we find strong evidence of tax-motivated profit shifting among corporate-group firms. We focus on corporate groups with both SMEs and non-SMEs. Columns 3 and 4 of Table 5 shows our estimation of Equation (4) with profit growth as a dependent variable. Our difference-in-difference analyses show that, following 2011, the profit growth of SMEs increases by 11.0 percentage points relative to that of non-SMEs.

We also find that transfer pricing is likely the primary channel for the profit shifting from non-SMEs to SMEs within the same corporate group. In Columns 5 and 6 of Table 5, we repeat the difference-in-difference analyses on the sample used in Column 4 but use input cost growth and debt growth as dependent variables instead. Our findings indicate that, since 2011, the growth of input cost of SMEs was 7.5 percentage points lower than that of non-SMEs, contradictory to the revenue growth which is not different between these two groups. The reduction is significant at the one-percent level. However, we do not find a significant change in the debt growth, implying that corporate groups do not change their financing policy in response to the SME tax incentive.

This result of profit shifting through cost reallocation is important because it underlines the importance of domestic transfer pricing—an issue which could be crucial for countries that allow preferential tax schemes for certain types of firms but so far has received little attention in the literature.

Our findings also indicate that the tax-motivated profit shifting is much more evident among small corporate groups. We perform heterogeneity analyses with respect to corporate group size, categorizing corporate groups into those with 10 or fewer firms (small) and those with more than 10 firms (large). Small corporate groups have 4,779 firms in total, while large corporate groups have 3,023 firms. The results are shown in Table 6. Since 2011, the profit growth of SMEs in small corporate groups increases by 12.8 percentage points relative to that of non-SMEs (Column 1). For large corporate groups, the interaction coefficient is positive but not significantly different from zero (Column 2). One possible explanation is that larger groups are monitored more closely by the regulator. Another explanation is that larger groups may have alternative mechanisms to respond to the tax incentive.

Finally, we further divide corporate groups into those with and without firms listed in

Table 5: Effects of the SME tax incentive on corporate-group firms

	Overall				Profit shifting channels	
	(1)	(2)	(3)	(4)	(5)	(6)
Dep Var =	Revenue growth	Revenue growth	Profit growth	Profit growth	Input cost growth	Debt growth
$Treat_{it} \cdot Post_t$	−0.003	−0.007	0.136***	0.110***	−0.075***	−0.049
	(0.008)	(0.009)	(0.033)	(0.034)	(0.023)	(0.036)
$Treat_{it}$	0.564***	0.615***	0.281***	0.281***	0.581***	0.252***
	(0.011)	(0.011)	(0.033)	(0.033)	(0.023)	(0.037)
$Post_t$	−0.051***	0.070	−1.108***	−2.728***	−0.422	0.699*
	(0.004)	(0.101)	(0.015)	(0.307)	(0.322)	(0.370)
Control	No	Yes	No	Yes	Yes	Yes
Sector-Year FE	No	Yes	No	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.056	0.101	0.167	0.167	0.133	0.063
Observations	97,597	97,562	44,222	44,222	38,915	29,514
Number of firms	8,635	8,633	7,804	7,804	6,855	6,366

Note: This table presents the estimated impacts of the SME tax incentive on corporate-group firms. $Treat_{it}$ is a dummy variable that equals one for firms that (1) have lagged revenue under the SME threshold and (2) have never had revenue above 30 million baht in the period from 2011 to the respective year, and equals zero for firms with lagged revenue above the SME threshold. $Post_t$ is a dummy variable that equals one for the years including and after 2011. The control variable is lagged value of total assets in log form. Standard errors are heteroscedasticity-robust and clustered at firm level. Numbers in parentheses indicate standard error. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.
Source: Authors' estimate.

the Stock Exchange of Thailand (SET). Being listed in the stock exchange is likely to indicate better corporate governance and supervision. We find that, among small corporate groups, the tax-motivated profit shifting is concentrated among those without any listed firms (Columns 3). We find no significant results for the groups that are likely to have better corporate governance (Columns 4 and 6). The result highlights the role of corporate governance in curbing domestic transfer pricing.

7 Survival and Formation of Firms

Finally, we revisit the intended objective of the SME tax incentive, whether it helps relax financial constraints of SMEs. We consider two outcome variables: survival of existing firms and formation of new firms.

7.1 Survival of Existing Firms

To examine how the SME tax incentive affects the survival propensity, we focus on firms around the threshold and perform the difference-in-difference estimation. We define our treatment group to be firms that satisfied the following conditions: (1) have lagged revenue in the range of 10–30 million baht and (2) have never had revenue above 30 million baht in the period from 2011 to the respective year. Control group includes firms with lagged revenue in the range of 30–50 million baht. Post includes years from 2011 onwards.

We define the survival variable, s_{it}, as a dummy variable equals one if firm i appears in the dataset in any year after year t and equals zero otherwise. To ensure that we have ample time to observe the existence of firms in the dataset, we include observations only up to 2014. The estimation period ranges from 2007 to 2014 so that we have the same number of years before and after the introduction of tax incentives. The equation can be written as

$$s_{it} = \alpha_1 Treat_{it} \cdot Post_t + \alpha_2 Treat_{it} + \alpha_3 Post_t + \epsilon_{it} \tag{5}$$

where s_{it} is defined as above, $Treat_{it}$ denotes a dummy variable that equals one for treated firms and equals zero for controlled firms, and $Post_t$ denotes a dummy variable that equals one for the years including and after 2011 and equals zero otherwise. Under the identification assumption that unobserved determinants of revenue growth (ϵ_{it}) do not change differentially on average across the treatment and control groups around the reform, the coefficient α_1 represent

Table 6: Heterogeneity analysis for corporate-group firms with respect to group size

	Group Size		Group Size x Governance			
$RevGrowth_{it}$	(1)	(2)	(3)	(4)	(5)	(6)
	Small	Large	Small x Non-listed	Small x Listed	Large x Non-listed	Large x Listed
$Treat_{it} \cdot Post_t$	0.128^{***}	0.042	0.118^{***}	0.367	-0.005	0.135
	(0.042)	(0.057)	(0.044)	(0.233)	(0.067)	(0.113)
$Treat_{it}$	0.216^{***}	0.365^{***}	0.193^{***}	0.566^{***}	0.370^{***}	0.348^{***}
	(0.046)	(0.048)	(0.048)	(0.189)	(0.057)	(0.094)
$Post_t$	-3.132^{***}	-1.757^{***}	-3.140^{***}	-0.104^{***}	-2.655^{***}	-2.176^{***}
	(0.487)	(0.279)	(0.495)	(0.001)	(0.517)	(0.336)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.377	0.409	0.381	0.535	0.412	0.486
Observations	26,090	18,122	24,297	1,793	11,772	6,350
Number of firms	4,779	3,023	4,441	338	1,992	1,031

Note: This table presents the estimated impacts of the SME tax incentive on profit growth of corporate-group firms. $Treat_{it}$ is a dummy variable that equals one for firms that (1) have lagged revenue under the SME threshold and (2) have never had revenue above 30 million baht in the period from 2011 to the respective year, and equals zero for firms with lagged revenue above the SME threshold. $Post_t$ is a dummy variable that equals one for the years including and after 2011. The control variable is lagged value of total assets in log form. Standard errors are heteroscedasticity-robust and clustered at firm level. Numbers in parentheses indicate standard error. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.
Source: Authors' estimate.

the causal effect of the SME tax policy on the survival propensity.

Our difference-in-difference analyses illustrate that the tax incentives enhance firm survival. As shown in Table 7, the probability that firms below the SMEs threshold survive until the next year was 0.2 percentage points higher than those above the threshold after the 2011 introduction of the tax incentives. The finding is robust across various specifications.

7.2 Formation of New Firms

To examine how the SME tax incentive affects formation of firms, we aggregate the firm-level data into the 4-digit ISIC industry level, compute the growth of firm registration and again perform the difference-in-difference estimation. We then categorize industries into two groups based on the extent to which firms engage with other firms (Business-to-Business or B2B transactions). Firms that heavily engage in B2B are likely to find it is harder to misreport sales and are therefore benefit more from the SME tax incentive once it is introduced. On the other hand, the opportunities to misreport revenue are relatively greater for firms that sell mostly to consumers (Business-to-Consumer or B2C transactions) due to the absence of the VAT enforcement mechanism. Firms in the B2C industries are thus benefit relatively less from the tax rate reduction. The estimation period ranges from 2004 to 2017. The equation can be written as

$$n_{st} = \alpha_1 Treat_s \cdot Post_t + \alpha_2 Post_t + \delta_s + \epsilon_{st} \tag{6}$$

where n_{st} denotes the growth of new firm registration of sector s in year t, $Treat_s$ denotes a dummy variable that equals one for treated sectors and equals zero for controlled sectors, $Post_t$ denotes a dummy variable that equals one for the years including and after 2011 and equals zero otherwise, and δ_s denotes a dummy variable for sector s.

In the broad specification, the treatment group includes wholesale trade and manufacturing industries, while the control group includes retail trade and hospitality industries. For the narrow specification, we focus only on the trading sector: our treatment includes only wholesale trade while the control includes only retail trade.

As shown in Table 8, our findings indicate that, following the introduction of the SME tax incentive, new firm registration associated with the treatment sectors increased by 16.9 percentage points relative to the control sectors (Column 2). The increase is significant at the 0.1 level. The finding is consistent with the narrow specification that includes only the wholesale trade and retail trade industries (Column 4).

Table 7: Effects of the SME tax incentive on survival probability

	(1)	(2)	(3)	(4)	(5)	(6)
$Treat_{it} \cdot Post_t$	0.002^{**}	0.002^{**}	0.002^{**}	0.002^{*}	0.002^{*}	0.002^{*}
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
$Treat_{it}$	-0.005^{***}	0.000	0.000	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)
$Post_t$	-0.005^{***}	0.000	-0.000	-0.024^{***}	-0.098^{***}	-0.040
	(0.001)	(0.013)	(0.013)	(0.001)	(0.018)	(0.061)
Lagged total asset (log)		0.007^{***}	0.005^{***}		0.003^{***}	0.003^{***}
		(0.000)	(0.000)		(0.000)	(0.001)
Lagged current asset (log)			0.002^{***}			0.001^{***}
			(0.000)			(0.000)
Firm age			0.001^{***}			0.004
			(0.000)			(0.010)
Sector-Year FE	No	Yes	Yes	No	Yes	Yes
Sector FE	No	Yes	Yes	No	No	No
Firm FE	No	No	No	Yes	Yes	Yes
R-squared	0.000	0.004	0.005	0.009	0.017	0.017
Observations	469,652	466,209	466,156	469,652	466,209	466,156
Number of firms	137,191	136,327	136,309	137,193	136,327	136,309

Note: This table presents the estimated impacts of the SME tax incentive on firm survival probability. The dependent variable is a dummy variable for firms' survival. $Treat_{it}$ is a dummy variable that equals one for firms that 1) have lagged revenue in the range of 10–30 million baht and 2) have never had revenue above 30 million baht in the period from 2011 to the respective year, and zero for firms with lagged revenue above 30 million baht but not over 50 million baht. $Post_{it}$ is a dummy variable that equals one for the years including and after 2011. Standard errors are heteroscedasticity-robust and clustered at firm level. Numbers in parentheses indicate standard error. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.
Source: Authors' estimate.

Table 8: Effects of the SME tax incentive on new firm registration

	Broad		Narrow	
	(1)	(2)	(3)	(4)
$Treat_s \cdot Post_t$	0.156*	0.169*	0.197	0.234*
	(0.090)	(0.092)	(0.130)	(0.129)
$Post_t$	−0.337***	−0.735***	−0.370***	−0.536*
	(0.086)	(0.124)	(0.099)	(0.290)
Year FE	No	Yes	No	Yes
Sector FE	Yes	Yes	Yes	Yes
R-squared	0.016	0.130	0.036	0.235
Observations	2,962	2,962	563	563
Number of sectors	294	294	51	51

Note: This table presents the estimated impacts of the SME tax incentive on new firm registration. The dependent variable is growth of new firm registration. $Treat_s$ is a dummy variable that equals one for sectors that mainly engage in B2B transactions and equals zero for sectors that mainly engage in B2C transactions. $Post_t$ is a dummy variable that equals one for the years including and after 2011. For Broad specification, Treat includes wholesale and manufacturing, while control includes retail and hospitality. For Narrow specification, Treat includes wholesale, while Control includes retail. Standard errors are heteroscedasticity-robust and clustered at industry level. Numbers in parentheses indicate standard error. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.
Source: Authors' estimate.

8 Conclusion

We use the introduction of the SME tax incentive in Thailand in 2011 to study the effects of size-dependent tax policies on firms behaviors. We find that firms respond strongly to this tax incentive and that the responses are heterogeneous across sectors and ownership structures.

Among stand-alone firms, we observe a sharp bunching just below the threshold, which did not exist before the incentive was introduced. In addition, after the incentive was introduced, the revenue growth of stand-alone firms below the SME threshold was significantly lower than the growth of those above the threshold. We also find that the bunching patterns are more prominent among firms with positive profit where the SME tax incentive produces an immediate benefit and more evident in the sector whose revenues are relatively easy to adjust or manipulate. Our study provides policy implications on the unintended consequences of SME incentive measures.

We do not observe any bunching among corporate-group firms. Instead, we find that after the incentive was introduced, the profit growth of SMEs increases significantly faster than that of non-SMEs, suggestive evidence of tax-motivated profit shifting among corporate-group firms. We find that transfer pricing is a likely primary channel and that the profit shifting is

more evident among small corporate groups with relatively weak governance. Our study thus contributes to the limited literature on the impacts of SME incentives and corporate governance on domestic transfer pricing. In addition, the findings in this paper shed some light on the tax-avoidance behavior of firms in developing countries where corporate groups are common and corporate governance is weak. The study therefore has an important policy implication on the design of better incentive-compatible SME-supporting policies.

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